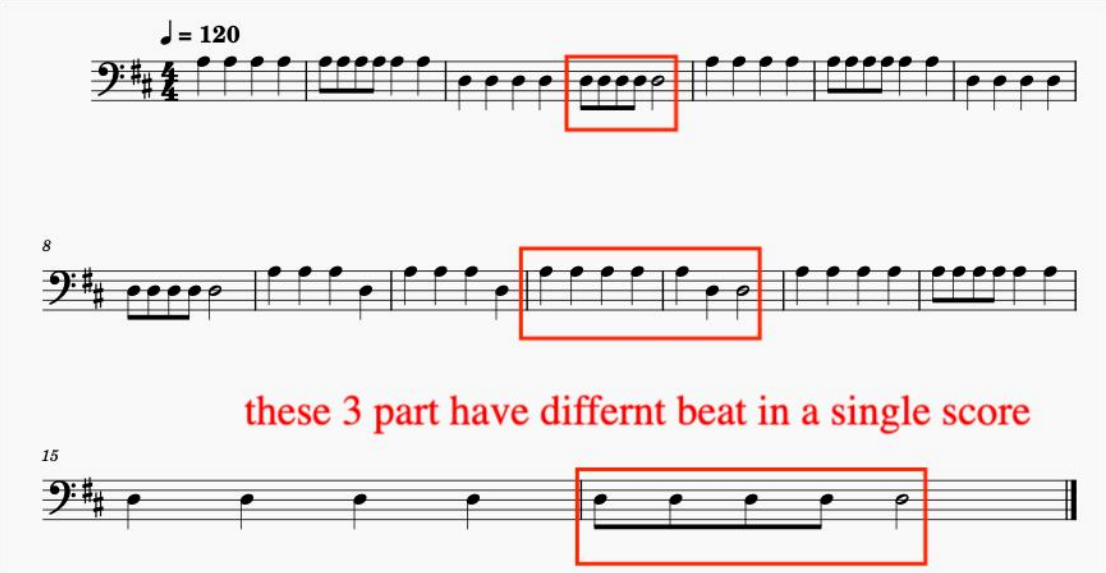
Normalized Inter-Onset Interval (IOI)

1. Folders

ioi_fold.py perform folder wide midi file VS audio file IOI scoring

ioi-2.py perform single ioi scoring

keys: normal ioi will include time that at different beat and treat them all as a single beat result in inaccuracy result. for these ioi-2 and ioi-folder will extra the different beat and treat them separately, so each time perform a ioi is based on the same beat that would have a more acculturate result.



2.Commands

command for ioi_fold.py

```
python ioi_fold.py \
  --midi_dir "Midi" \ (Midi file folder)
  --audio_dir "Robot" \ (Robot performing folder)
  --lib "ioi-2.py" \
  --auto-allowed \
  --ignore-long-beats 8.0 \
  --save_plot aggregated_ioi.png
```

command for ioi-2.py

```
python ioi-2.py Midi/long-long-ago.mid  Robot/long-long-ago.mp3 --plots --auto-allowed
```

2. Example Result based on old file performance

2.1 ioi-2 result:

=== MIDI↔AUDIO IOI CONSISTENCY ===
file_midi Midi/allegro.mid
file_audio Robot/allegro.mp3
instrument_program 42
midi_ioi_beats_stats {'n': 71, 'mean': 0.8732394366197183, 'median'...\
(found 71 ioi, mean beat 0.87)
audio_ioi_beats_stats {'n': 87, 'mean': 1.539071353608203, 'median':...
ratio_all_stats {'n': 50, 'mean': 1.7682288254516823, 'median'..
(match 50 between midi and audio; audio ioi is 1.768 longer than midi)
matched_pairs 50
tempo_est_bpm 92.285156
dtype: object

Per-category (played / expected, 1.0 ideal):

	expected_beats	count	mean	median	std	cv
0	0.5	20	2.100297	1.060082	2.430351	1.157146
1	1.0	27	1.551733	1.128896	0.999408	0.644059
2	2.0	3	1.502901	1.508949	0.496179	0.330147

(based on each beat-- how each beat performs)

2.2lil-fold result

==== SINGLE DATASET-WIDE AVERAGE COMPARISON ====
Pairs processed: 5/5
Total ratio samples: 458
Global mean ratio: 1.23065
Global median ratio: 0.974412
Global std dev: 1.30665
Mean of per-pair metric (e.g., score/ratio_mean): 1.11503
(overall performance)

4. Graphs

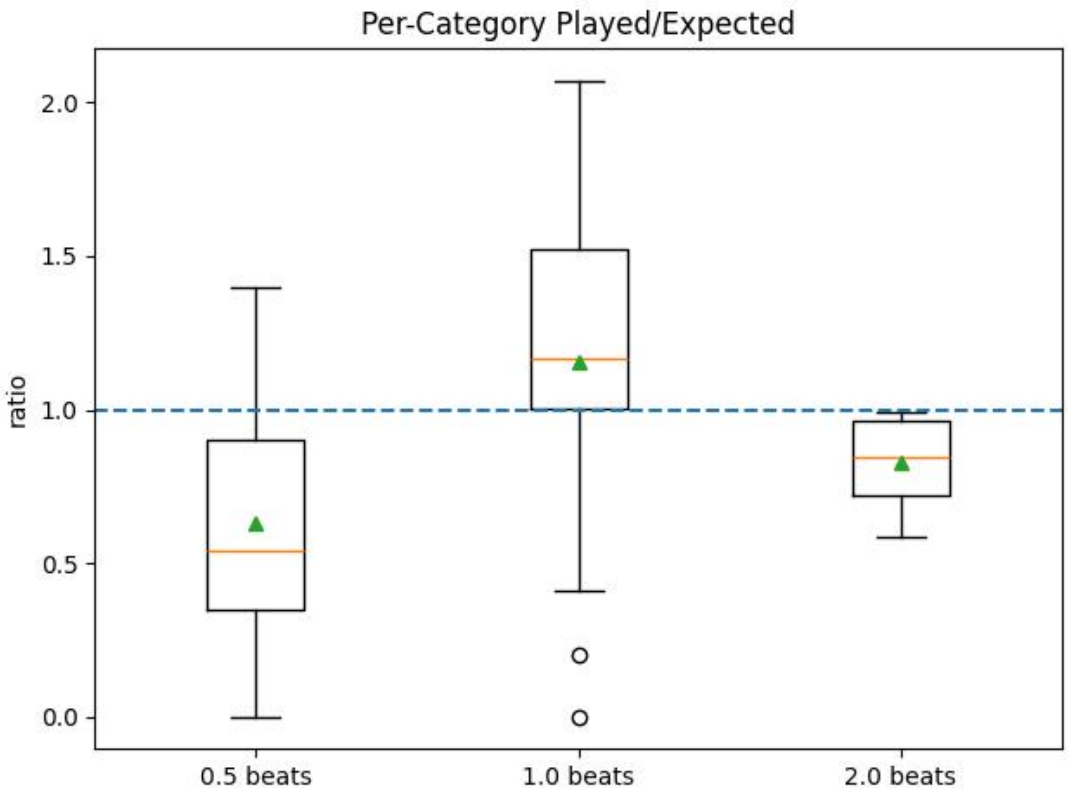


Figure1: ioi-2 graph for single comparison on onset for different beats

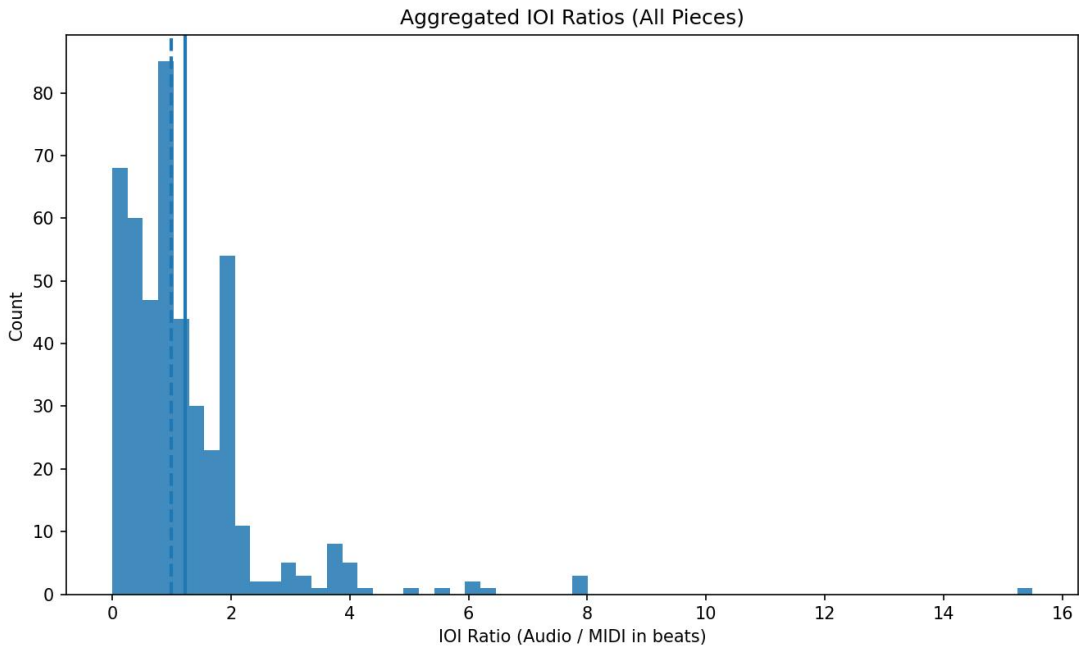


Figure2: ioi-fold graph for all pieces match on original file